

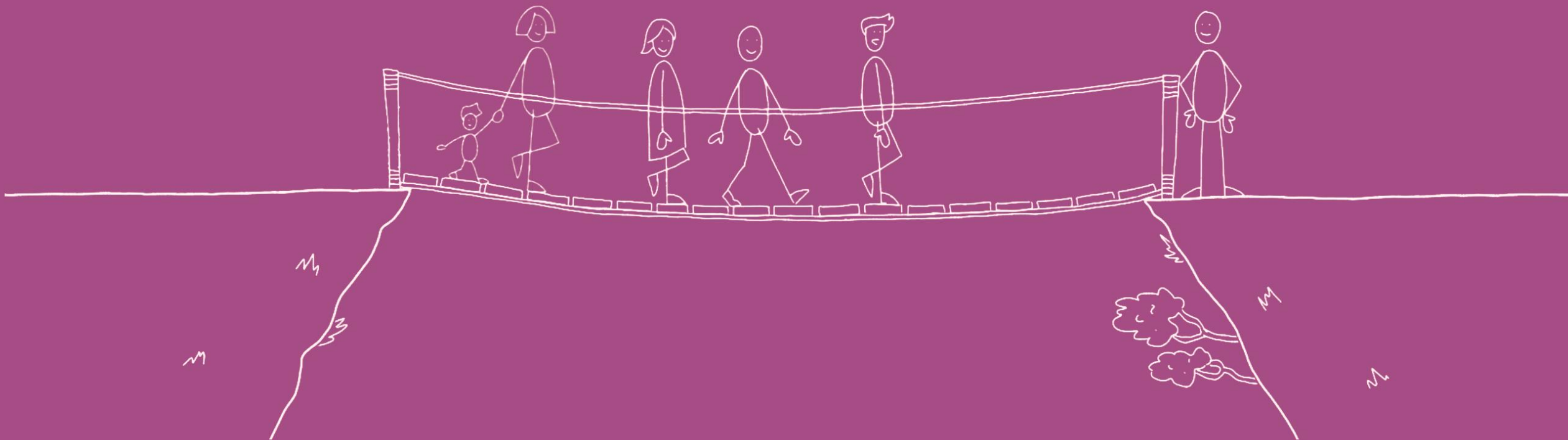


WELCOME

inpeek
empowering and connecting people



WHAT IS INPEEK?



About inpeek



FOUNDATION

February 2009



OFFICES

Bern – Headquarter,
Zürich,
Darmstadt (D)



EMPLOYEES

75



SAP AND MICROSOFT PARTNER

SAP Gold Partner,
SAP Recognized Expertise
SAP HANA und
SAP User Experience,
Microsoft Partner



PROJECT PORTFOLIO

Over 500 successful projects and
mandates.

inpeek AI Unit

We accompany our customers from the idea over conception to final implementation. With in-depth SAP and AI expertise as well as complete integration, we create added value.



Fabian Hüni
Head of AI
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Nicolas Wyss
Junior AI Engineer
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Fabian Hüni

- At inpeek for **3.5 years**
- Head of the newly established AI Unit
- Particularly interested in Generative AI, LLM, Agents, Graphs and browser-based AI

Education

- MSc AI in Medicine, University of Bern | expected 2026
- BSc in Computer Science, University of Bern | 2024
- Apprenticeship as IT specialist | 2020



Applied AI Journal Club – powered by inpeek



Analysis of Applied AI papers, **relevant to the business world**



Papers are **read in advance** and **discussed during the meetup**



Inpeek AG
Bogenschützenstrasse 9
3008 Bern



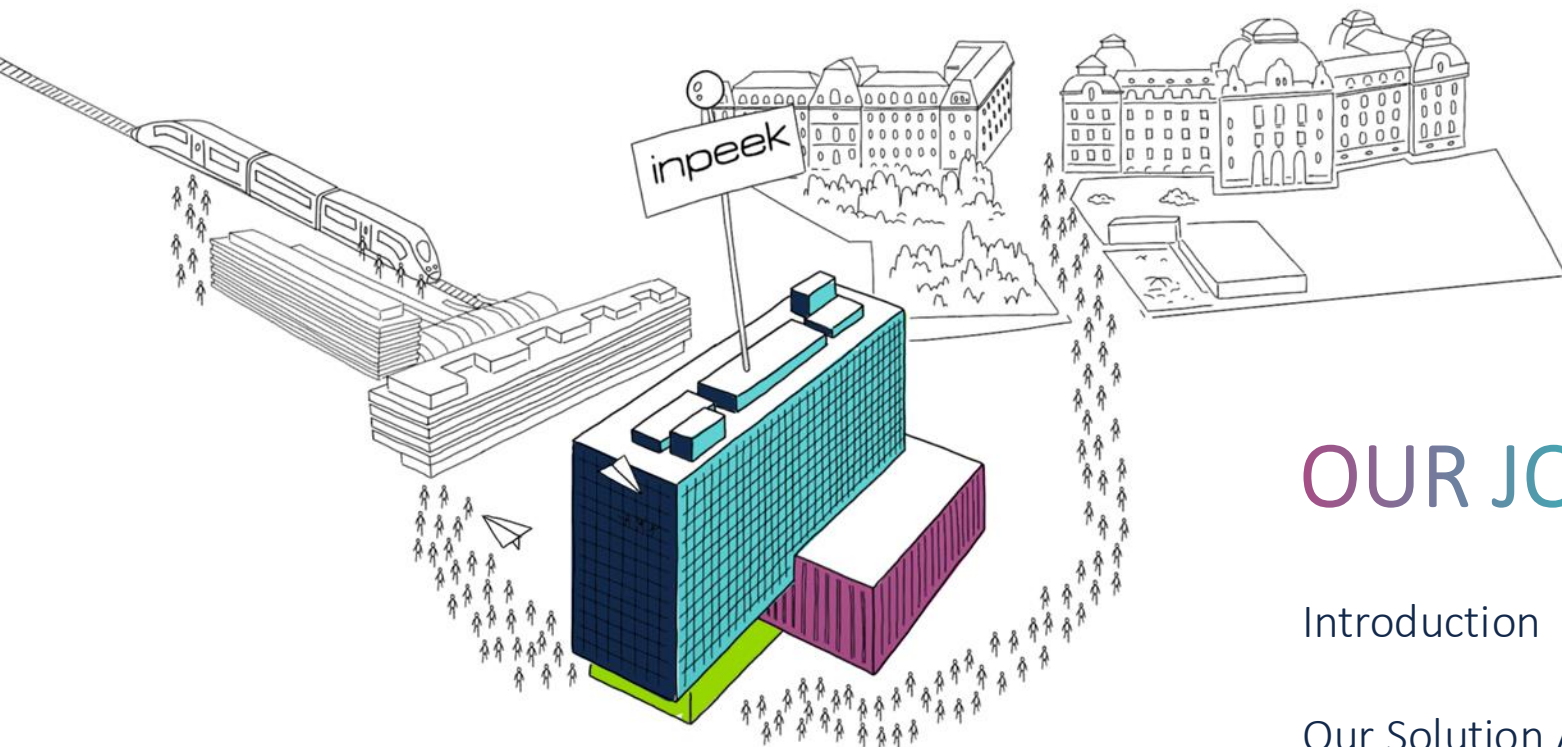
24th February 17:30-18:30
*Small Language Models for
Efficient Agentic Tool Calling*



GRAPH-BASED RAG FOR STRUCTURED DATA

inpeek
empowering and connecting people

26. Januar 2026
Fabian Hüni



OUR JOURNEY

Introduction

Our Solution Approach

Technical Deep Dive

Evaluation & Observability

Conclusion

INTRODUCTION

Graph-based RAG for Structured Data | January 26, 2026 inpeek



Summarix

Summarix is our tool for managing employees, projects, skills, or customers.



Available on
SAP Store

Reference Tool

Profile

Mitarbeitende



Skills



Dokumentation

Kunden & Projekte

Kunden



Projekte



Kontaktpersonen



Segmentierung

Branchen



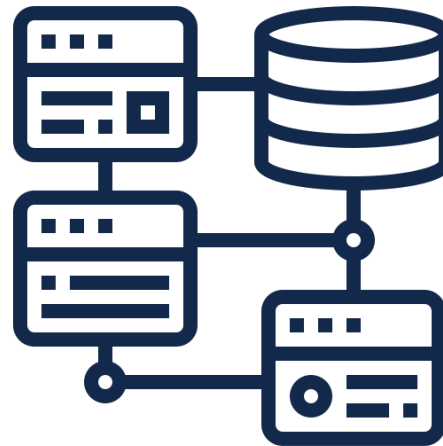
Units



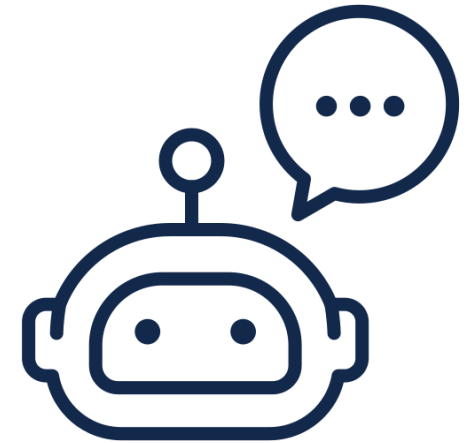
Challenges



Inefficient search for
reference projects

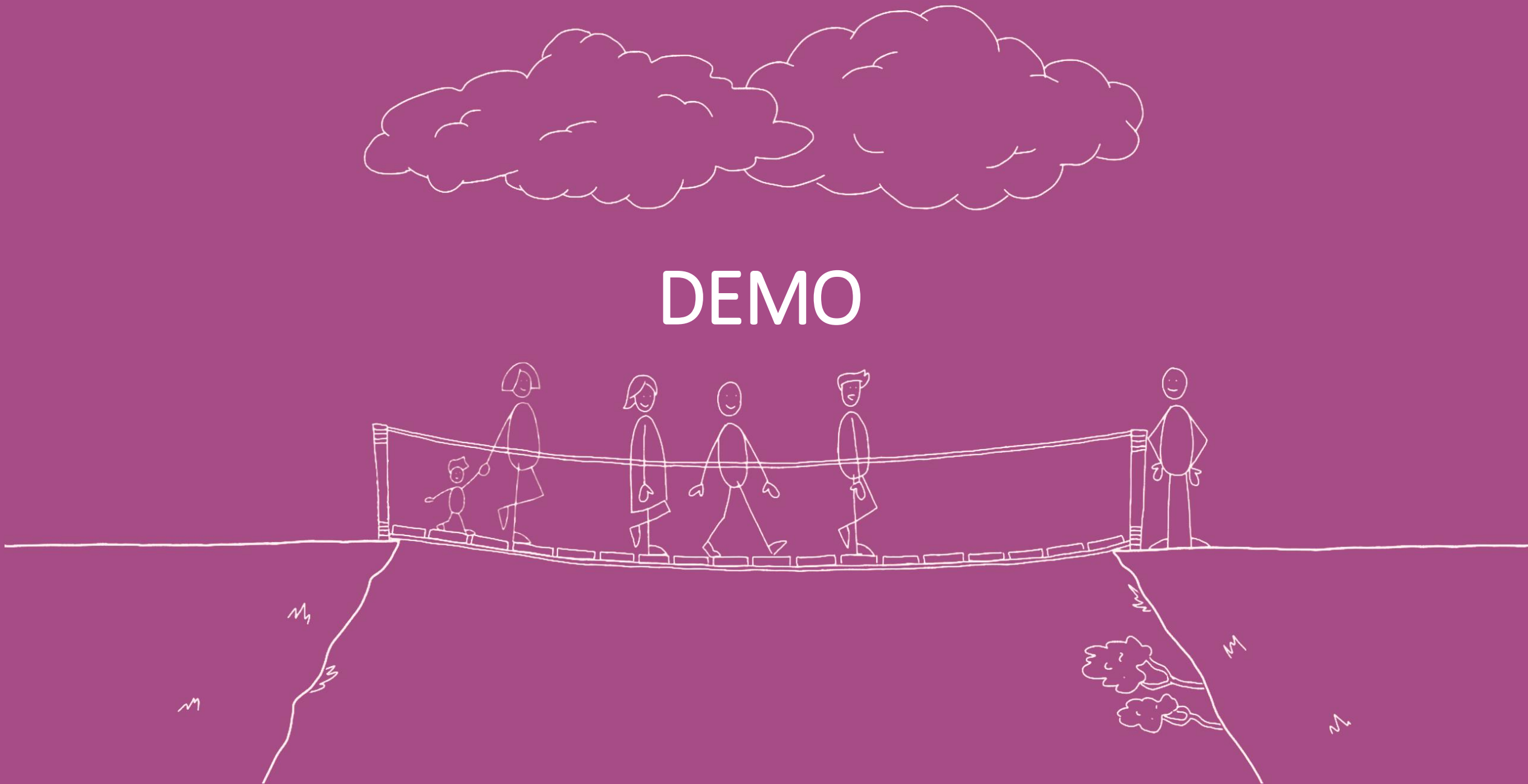


Summarizing projects
and profiles



Time-consuming search
for colleagues

DEMO



Reference Tool

Profile

Mitarbeitende



Skills



Dokumentation

Kunden & Projekte

Kunden



Projekte



Kontaktpersonen



Segmentierung

Branchen

Units

Summy



No messages yet.

Type a message

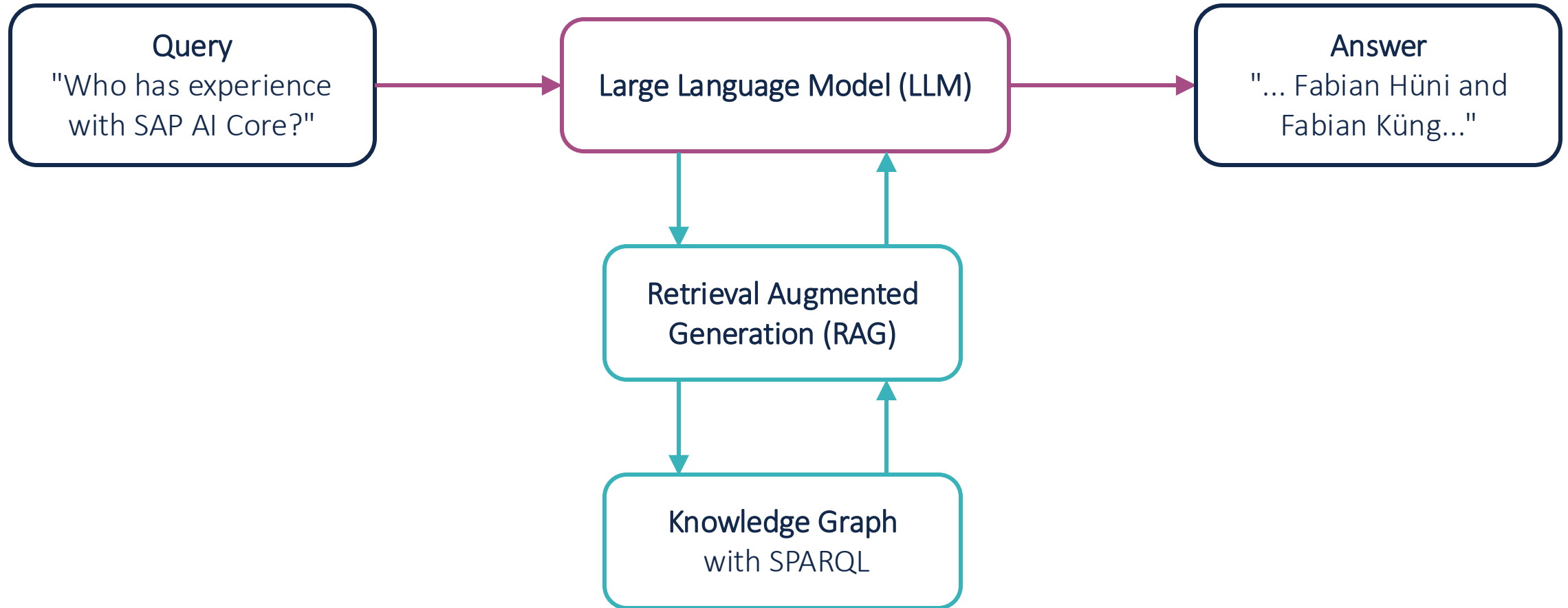


SOLUTION

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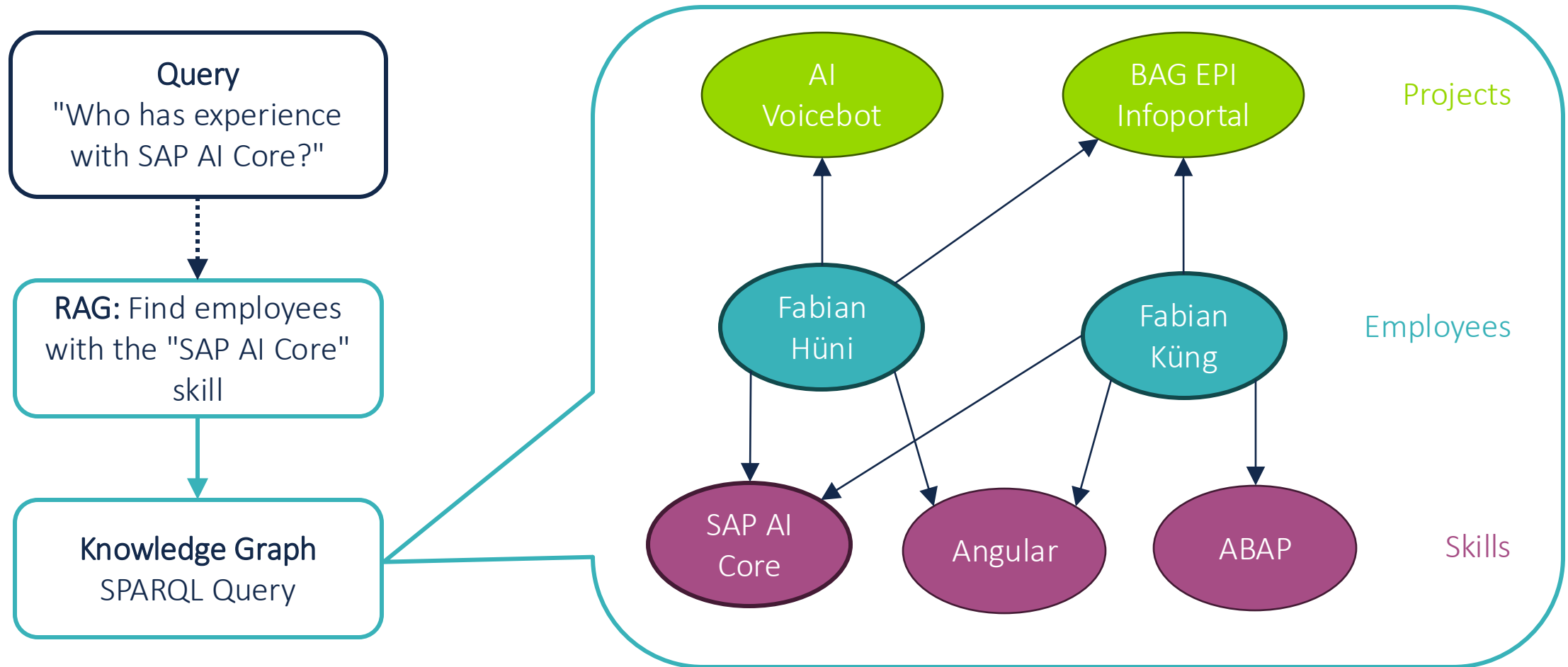
Retrieval Augmented Generation (RAG)



But what about this Knowledge Graph?



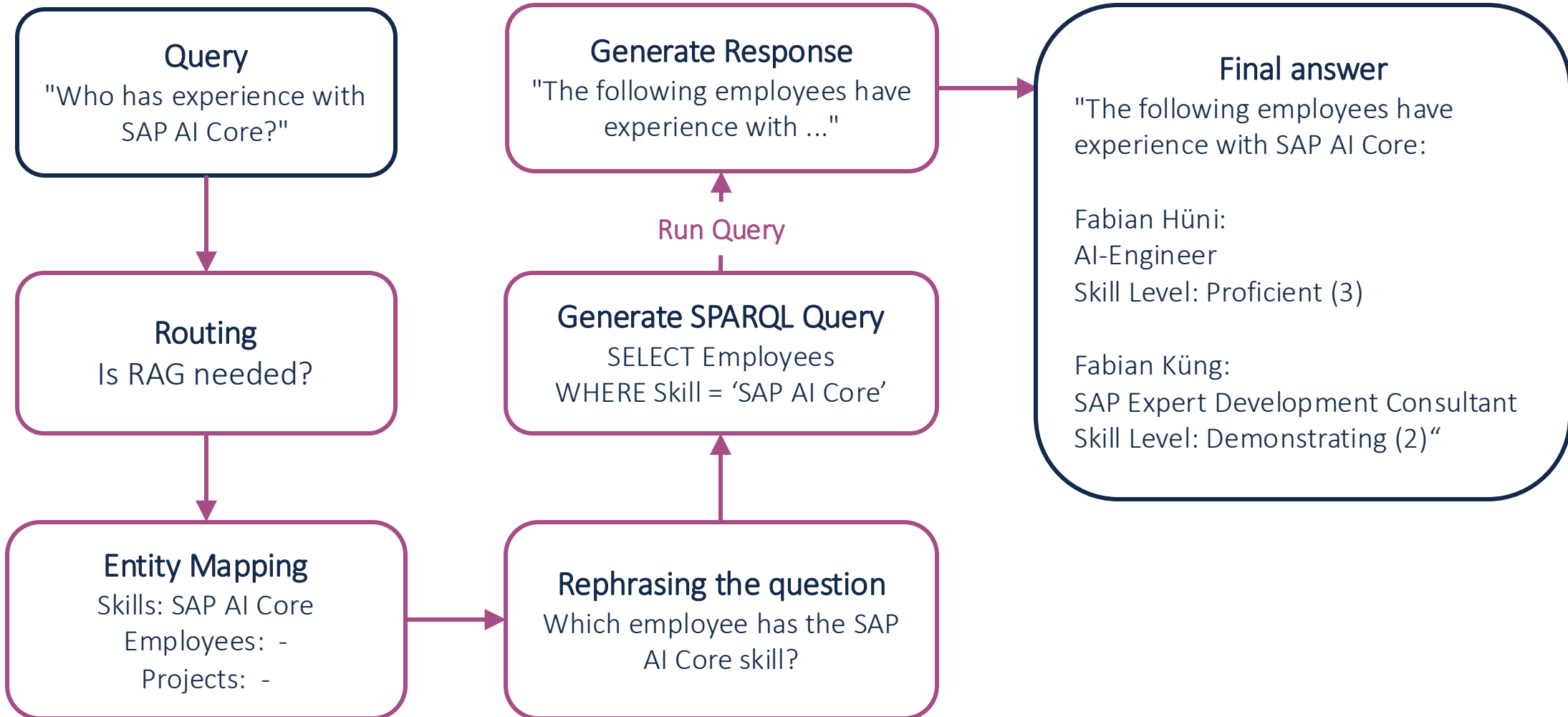
RAG with Knowledge Graphs



How does the retrieval process work?



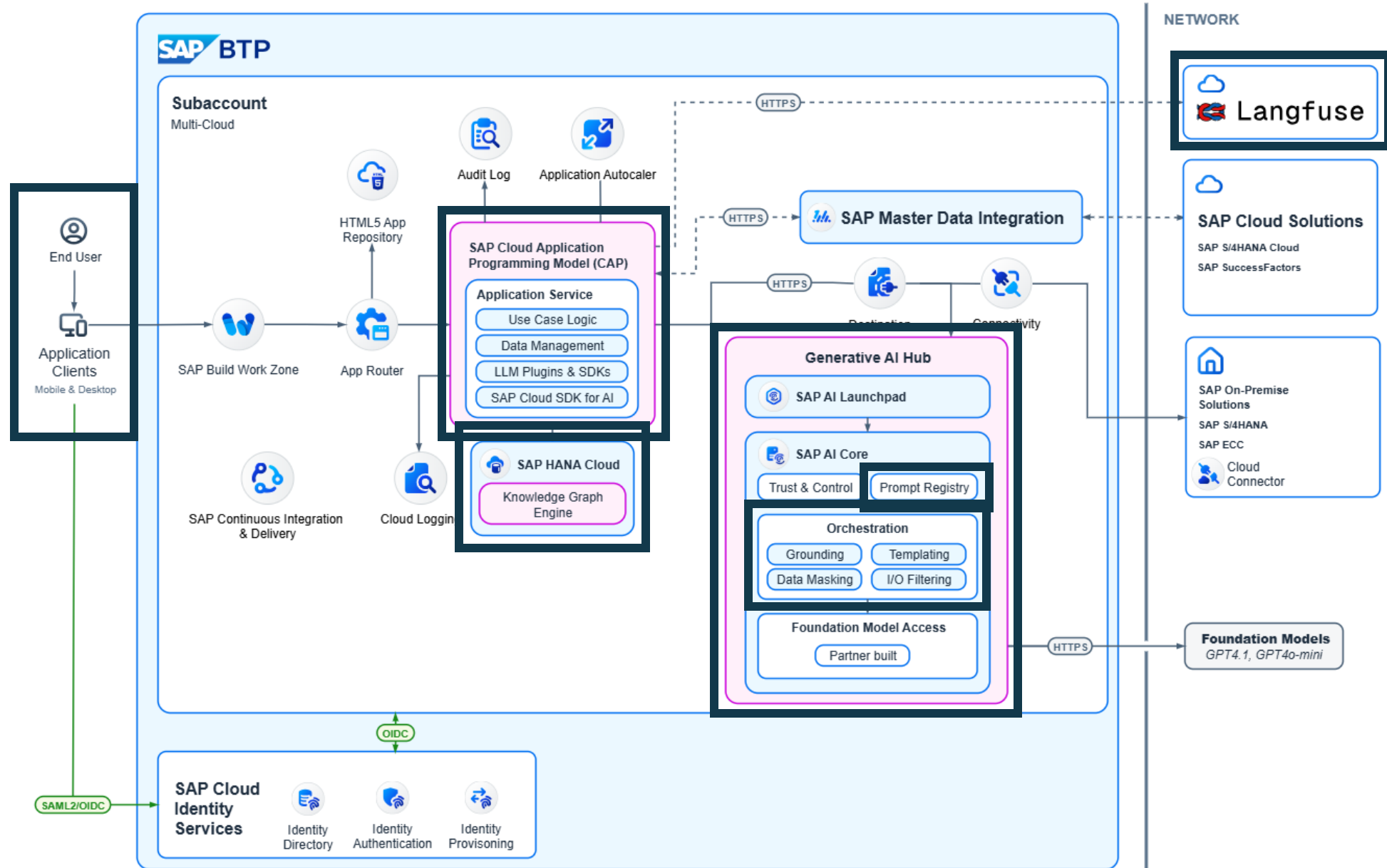
Agentic Workflow



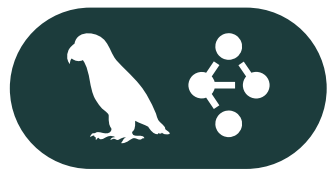
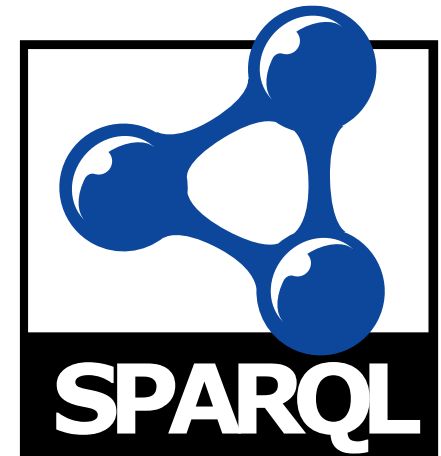
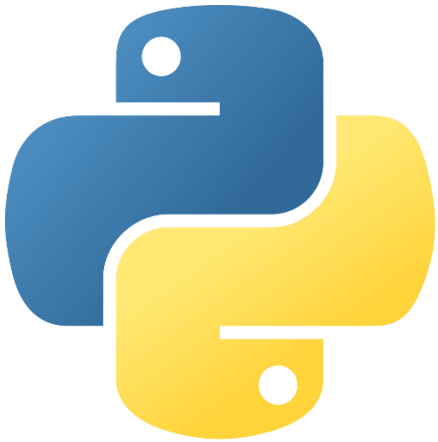
TECHNICAL DEEP DIVE

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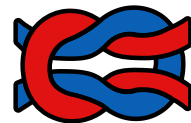




Tech Stack



LangGraph



Langfuse

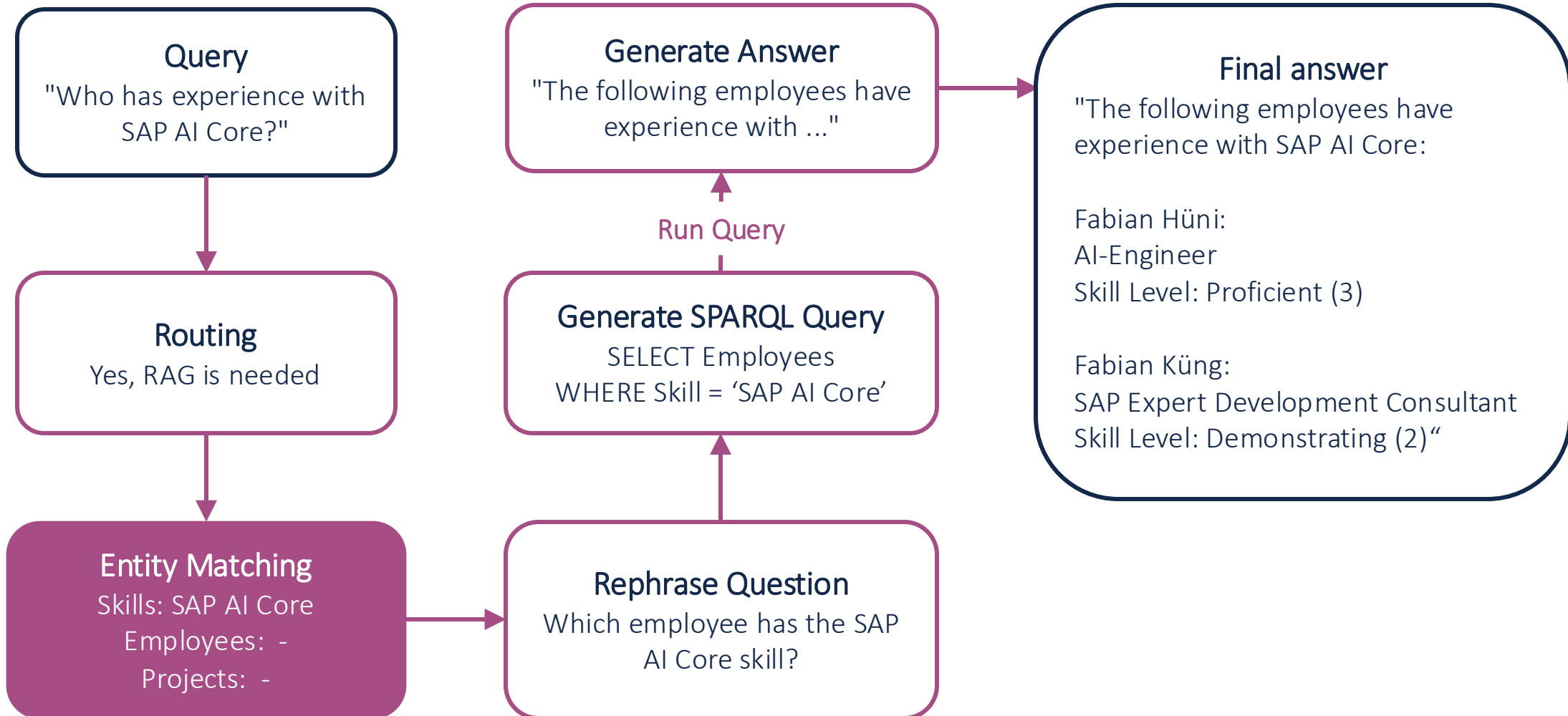
What have been the main challenges?



Context Engineering

What information is needed
when and in what form
to successfully execute the task?

Recap: Agentic Workflow



Entity Matching - Challenge

Entity	Possible inputs	In the system
Skill	SAP BTP SAP Business Technology Platform BTP Business Technology Platform	SAP Business Technology Platform (BTP)
Employee	Fabian Hüni Fabian Hüni	Fabian Hüni
Skill / Job Title	Project Manager Project Manager Project management	Project Manager
Customer	SBB SBB AG Swiss Federal Railways Ltd Schweizerische Bundesbahnen (SBB)	SBB AG

Entity Matching - Solution

LLM-Prompt

- Spelling correction
- Transformation in singular
e.g. "Projektleiterinnen" → "Projektleiterin"
- Generation of multiple variants
- **Stemming** (reduction to word stem),
e.g. "Projektleiterin" → "Projektleit"

Coding

- Split for spaces
e.g. "SAP BTP" → "SAP", "BTP"
- Regex with case-insensitive matching per search term
- Linking the search terms
AND / OR depending on the entity

Example

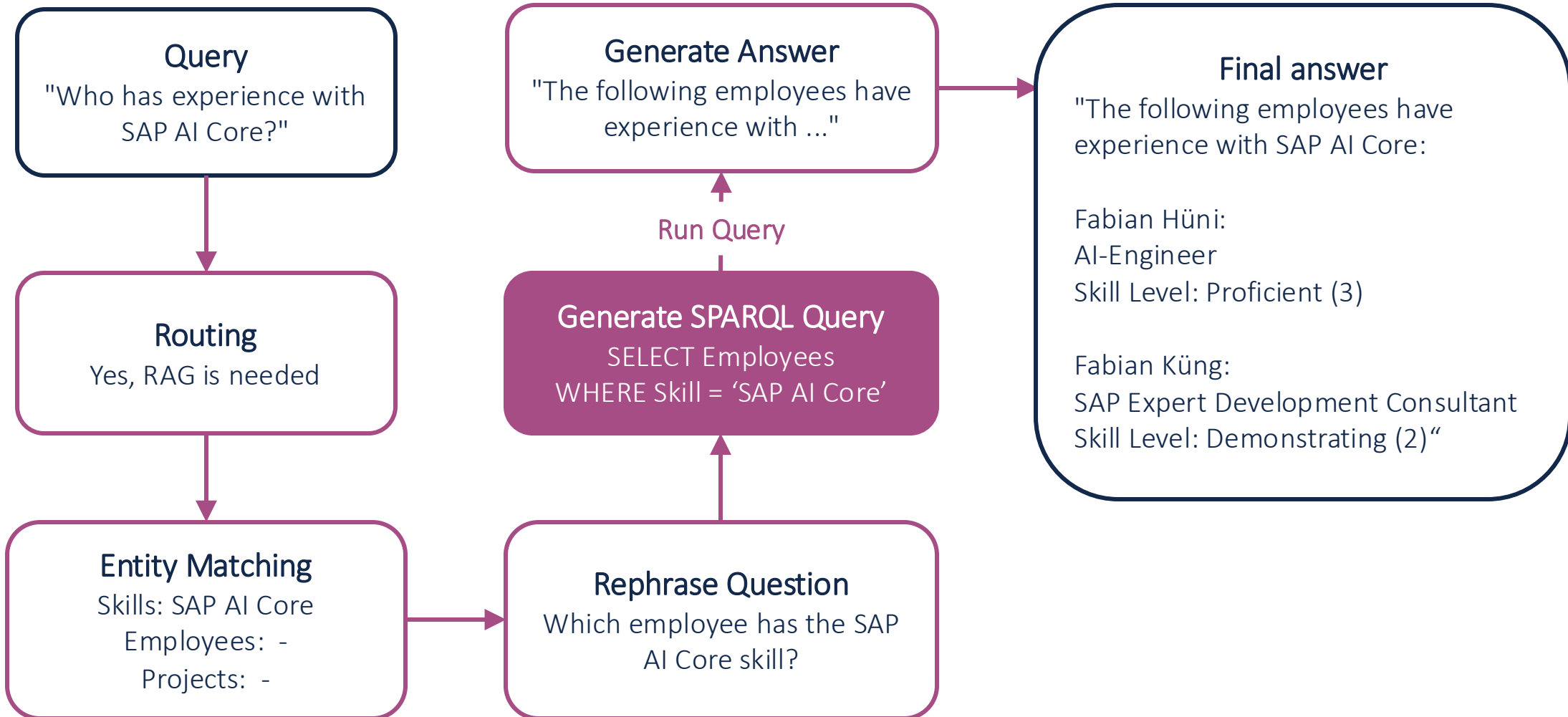
"SAP BTP" becomes

`FILTER regex(STR(? skillName), "SAP", "i").`

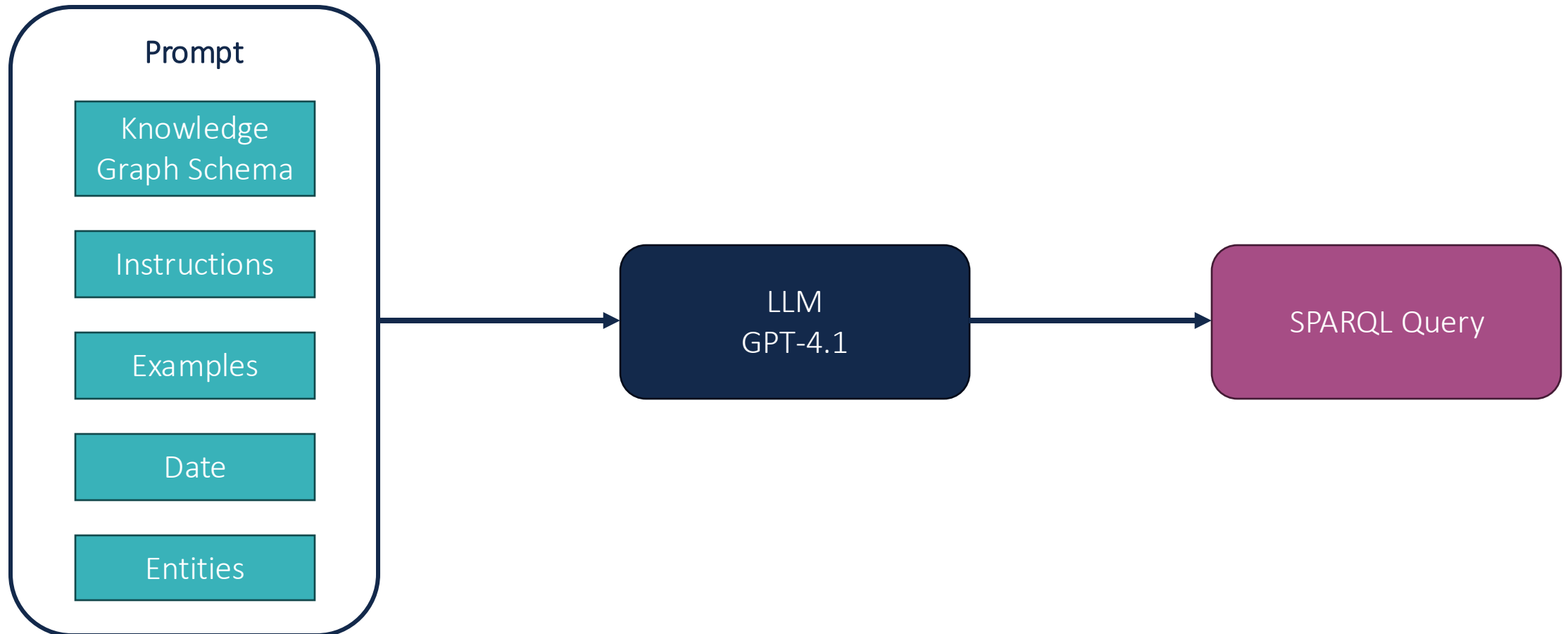
`FILTER regex(STR(? skillName), «BTP", "i").`

= All skills that have "SAP" and "BTP" in their names

Recap: Agentic Workflow



SPARQL Query Generation



EVALUATION & OBSERVABILITY

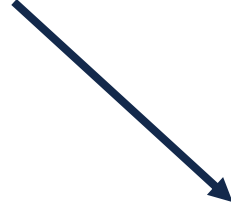
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Evaluation & Observability

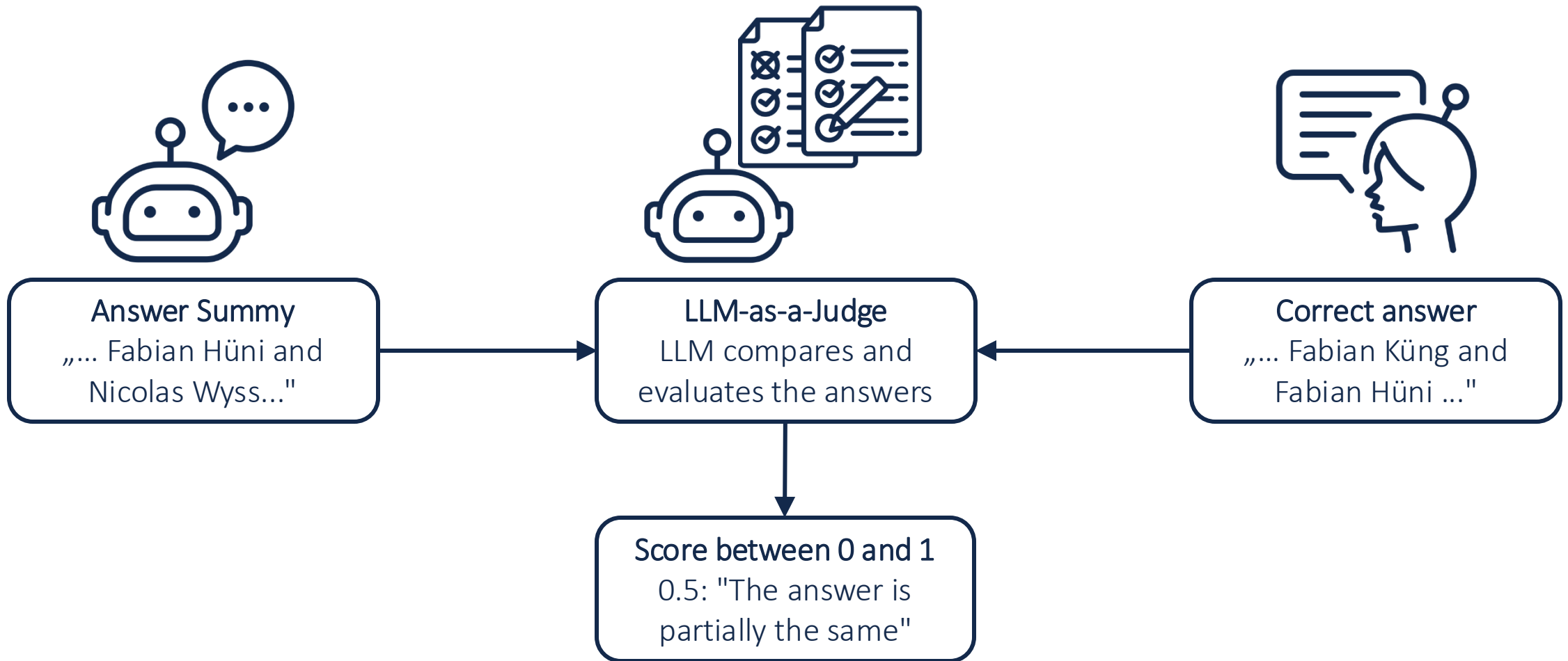


By how many percentage points has the **performance of** our system changed as a result of the improvement X?

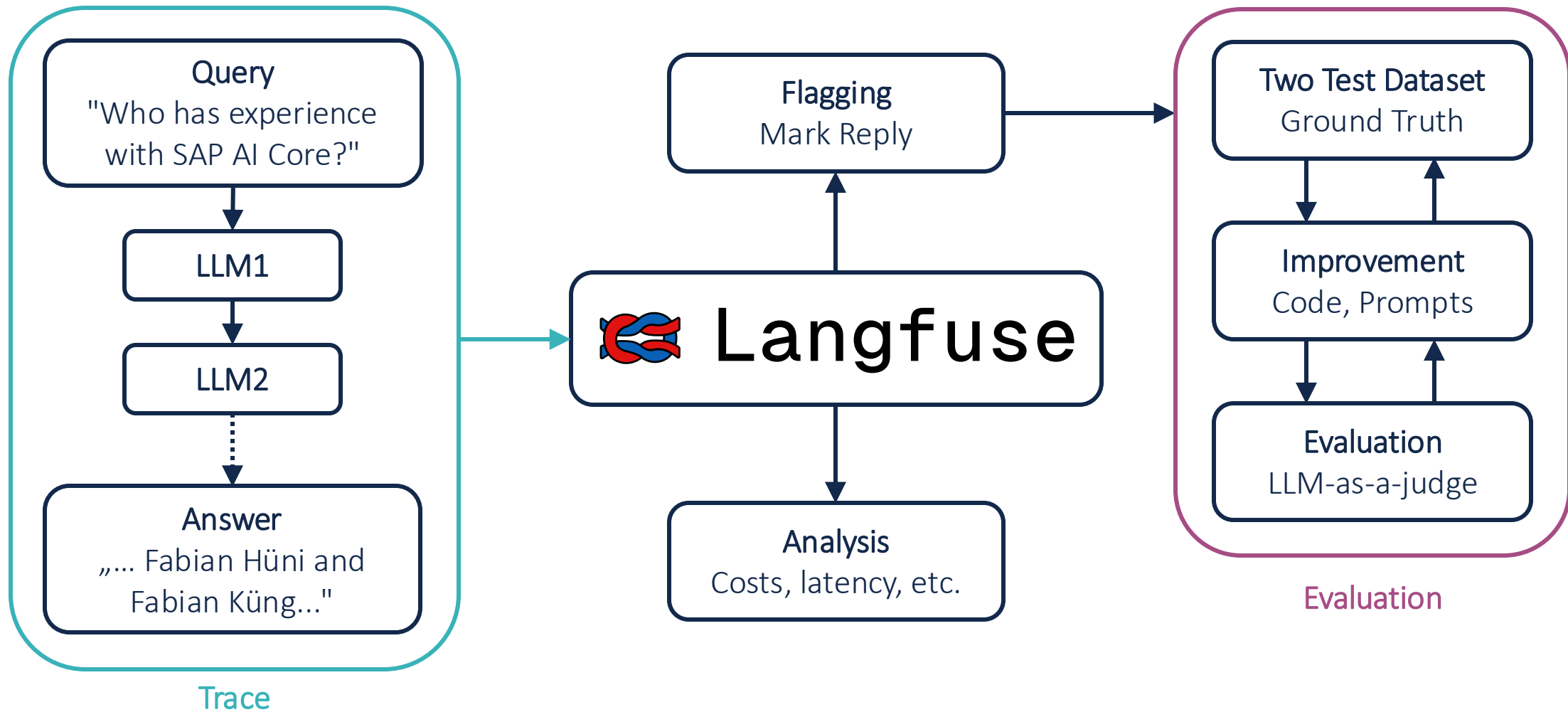


Traceability of individual runs of the Agentic AI workflow
(which actions were carried out and what results were produced).

LLM-as-a-Judge



Evaluation & Observability



Langfuse



Langfuse v3.112.0

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

LLM-as-a-Judge

Human Annotation

Datasets

Upgrade

Settings

Support

F

Fabian

fabian.hueni@inpeek...

inpeek-dev

Hobby

reference-tool

Datasets

Dataset reference-tool-eval

New dataset run

1 active evaluators

Charts

2

K

J

Runs

Items

Latency (s)

Correctness (eval)

Filters

Columns 13/14

☐	Name	Description	Run Items	Latency (avg)	Total Cost (avg)	Total Cost (sum)	Run-Level Scores	# corrections (api)	# Correctness (eval)	# sparql_errors
☐	issues-0.3-3 (basic sq...		26	6.05s	\$0.00	\$0.00		0	0.7731	0
☐	issues-0.3-2 (added st...		26	11.38s	\$0.00	\$0.00		0	0.8769	0
☐	issues-0.3-1 (updated ...		26	13.97s	\$0.00	\$0.00		0	0.8269	0
☐	issues-0.2-16 (new ont...		26	12.48s	\$0.00	\$0.00		0	0.7115	0
☐	issues-0.2-15 (gpt-4.1 s...		26	13.85s	\$0.00	\$0.00		0	0.7385	0
☐	eval-0.2-14 (industry s...		26	7.47s	\$0.00	\$0.00		0	0.8192	0
☐	eval-0.2-12 (sparql enti...		26	6.78s	\$0.00	\$0.00		0	0.8385	0
☐	eval-0.2-12 (gpt-4-1-na...		24	7.90s	\$0.00	\$0.00		0	0.6708	0
☐	eval-0.2-11 (Entity Sear...		26	7.18s	\$0.00	\$0.00		0	0.8019	0
☐	eval-0.2-10 (Employee ...		26	8.46s	\$0.00	\$0.00		0	0.8173	0
☐	eval-0.2-9 (After Fixing)		26	4.83s	\$0.00	\$0.00		0	0.8154	0
☐	eval-0.2-8 (Added Exa...		26	5.53s	\$0.00	\$0.00		0	0.7462	0
☐	eval-0.2-7 (Added Attr...		26	8.46s	\$0.00	\$0.00		0	0.4250	0
☐	eval-0.2-6 (Description...		26	7.24s	\$0.00	\$0.00		0	0.5885	0
☐	eval-0.2-5		26	7.74s	\$0.00	\$0.00		0	0.3423	0

Rows per page

50

Page

1

of 1

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CONCLUSION

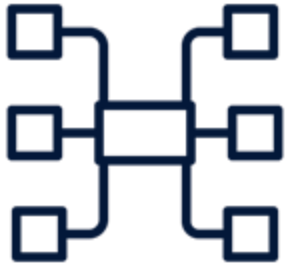
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Is the effort for a knowledge graph worth it?



Lessons Learned



Entity Mapping



Evaluation is Key



LLM-as-a-Judge



Context Engineering

QUESTIONS



THANK YOU



Fabian Hüni



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